

Expired by Climate

Do pharmaceutical stability test conditions still cover the climates they were built for? A 45-year audit using reanalysis and weather-station data

[Your name] · [Institution or city] · October 2026

Summary

Every medicine's shelf life is proven by storing it at a fixed test condition: 25 °C for temperate markets, 30 °C for hot ones. Those numbers were chosen decades ago to sit, with a safety margin, above the climate in which medicines are stored. We re-ran the regulators' own climate calculation on 1981–2025 data to see how much of that margin is left.

Data: 18 places — 10 from ERA5 reanalysis, 8 from NOAA weather stations; 1981–2000 vs 2006–2025.

Finding 1: the effective storage temperature (mean kinetic temperature) rose in all 18 places, by 0.35 to 1.33 °C; every 95% interval lies above zero.

Finding 2: after correcting a measured method bias, 7 of 18 places now average above the test temperature their medicines are approved at (Phoenix, Las Vegas, Dubai, Tucson, Miami, Tampa and Houston). On the US Gulf coast, inside an ICH region, Houston and Tampa crossed the 25 °C line between the two periods (Houston 24.68 → 25.41 °C; Tampa 24.96 → 25.67 °C), and New Orleans now exceeds it in 9 of the last 20 years.

Finding 3: in humid coastal cities (Chennai, Colombo, Yangon) the air already holds more water vapour than the 30 °C / 65% test chamber.

Control: Tokyo stays inside Japan's 25 °C condition, reproducing the only published re-check (Miura, 2024).

Why now. The International Council for Harmonisation (ICH) plans to adopt its consolidated stability guideline, Q1, in November 2026. This study offers a reproducible way to check, place by place, whether the climate assumptions behind its test conditions still hold.

Sections (in the order required by the Research Track): 1 Background & Problem Definition · 2 Literature Review · 3 Proposed Solution / Theoretical Framework · 4 Implementation Plan (with first results) · 5 References · 6 Prototype, Diagram & Visualization.

1. Background & Problem Definition

Medicines degrade through chemical reactions whose speed rises steeply with temperature and, for many solid products, with humidity. To prove that a medicine stays within specification for its whole shelf life, regulators require *long-term stability studies*: batches are stored for 12 months or more at a fixed condition and tested at intervals. The World Health Organization (WHO) specifies three long-term conditions — 25 °C / 60% relative humidity (RH), 30 °C / 65% RH and 30 °C / 75% RH — chosen according to the *climatic zone* in which the product will be marketed (WHO, 2018).

These numbers are not arbitrary. They were derived by turning weather records into a *mean kinetic temperature* (MKT): the single constant temperature that would cause the same total degradation as the real, fluctuating one (Haynes, 1971). Grimm (1998) set the hot-zone conditions with a safety margin above both these calculated climate values and temperatures measured in warehouses. The margin is the medicine's climate buffer.

The climate those calculations described has changed. If a region's effective temperature now exceeds the condition its medicines were tested at, every shelf life approved on that basis is optimistic by an amount that can be calculated from the Arrhenius equation. WHO itself warns that testing at conditions milder than the market's climate "could potentially lead to substandard products" (WHO, 2018).

Why it matters. Degraded medicines under-dose patients. For antibiotics, sub-therapeutic doses can select for resistance (Kaplan, Hamer & Shioda, 2025). The question is also timely: ICH is consolidating its stability guidelines into a single Q1 guideline, with adoption planned for November 2026 (ICH Q1 EWG, 2026).

Problem statement. Stability test conditions were fixed once, from late-twentieth-century climate data, and assigned country by country. There is no routine, quantitative check of whether today's climate still sits inside them. **Research question:** between 1981–2000 and 2006–2025, how much of the margin between each long-term test condition and the climate it represents has been consumed, and where has the climate already crossed it?

2. Literature Review

2.1 How the conditions were built

Haynes (1971) introduced mean kinetic temperature and used it to compute "worldwide virtual temperatures" for stability testing. Schumacher and Grimm used MKT and water-vapour pressure to divide the world into four climatic zones; ICH guideline Q1F cites Grimm's work as the basis of that classification (ICH, 2003b). Grimm (1998) derived 30 °C / 35% RH for hot-dry Zone III and 30 °C / 70% RH for hot-humid Zone IV, with a safety margin over calculated climate data and measured warehouse data. ICH Q1A(R2) (2003a) fixed 25 °C / 60% RH for the ICH regions (Europe, Japan, United States).

2.2 Revisions for the tropics

Zahn et al. (2006) computed temperature and vapour-pressure statistics from meteorological measurements at tropical sites and proposed explicit safety margins, supporting a split of Zone IV. ICH adopted 30 °C / 65% RH for Zones III/IV in 2003, but several hot and humid countries wanted a larger margin; ICH withdrew Q1F in 2006 and WHO introduced Zone IVa (30 °C / 65% RH) and Zone IVb (30 °C / 75% RH) (WHO, 2018). WHO's Eastern Mediterranean office used ERA-40 reanalysis to assign zones in its region (WHO EMRO, 2006). Taiwan derived its own conditions from 1991–1995 station data (Journal of Food and Drug Analysis, 1997).

2.3 Climate change and medicine quality

Only one modern re-check of the conditions against recent climate was found: Miura (2024) recomputed MKT for major Japanese cities from 1991–2020 data and found the ICH conditions still satisfied. Pericão et al. (2026) proposed revising the European Medicines Agency's storage-statement guideline to reflect climate variability, but did not recompute any zone. Kaplan, Hamer and Shioda (2025) call for research on how heat and humidity extremes affect medicine quality and link poor-quality medicines to antimicrobial resistance. Jenkins, Cancel and

Layloff (2022) used MKT to model shelf-life loss from temperature excursions in health programmes. A 2011 regulatory thesis suggested Zone II countries might need to reassess their conditions as the climate warms (Baik, 2011).

2.4 The gap

What is known	Limitation	What this study adds
Conditions derived from climate data via MKT and vapour pressure (Haynes 1971; Grimm 1998)	Derived once, from data before recent warming	A repeated, year-by-year re-derivation, 1981–2025
Single-country re-checks: Taiwan 1997, Eastern Mediterranean 2006, Japan 2024	No multi-decade, multi-country comparison	18 places across Asia, Europe and North America; two 20-year periods; same method
Zones assigned per country	Hotter regions inside a country are hidden	Sub-national places tested against their country's condition
Humidity drives degradation of many solid products	Vapour-pressure trends rarely tracked	Temperature and humidity margins measured separately
ICH Q1 is being finalised (Step 4 planned Nov 2026)	No public quantitative climate audit found	An open, reproducible audit method

Search method and calibration. About 25 searches (8 October 2026) of journal, regulator and author pages, using the terms *climatic zone*, *mean kinetic temperature*, *stability testing*, *climate change*, *Zone IVb*, *Grimm*, *Zahn* and *ICH Q1 revision*, 1971–2026. This is a scoping search, not a systematic review: "not found" means not found in these searches, not that no such work exists.

3. Proposed Solution / Theoretical Framework

Core idea. Treat a regulatory test condition as a *hypothesis about the climate* — "the effective temperature here stays below this number, with a margin" — and test that hypothesis with the regulators' own method on today's data. The formula is not new; what is new is applying it repeatedly over time, below country level, and to temperature and humidity separately.

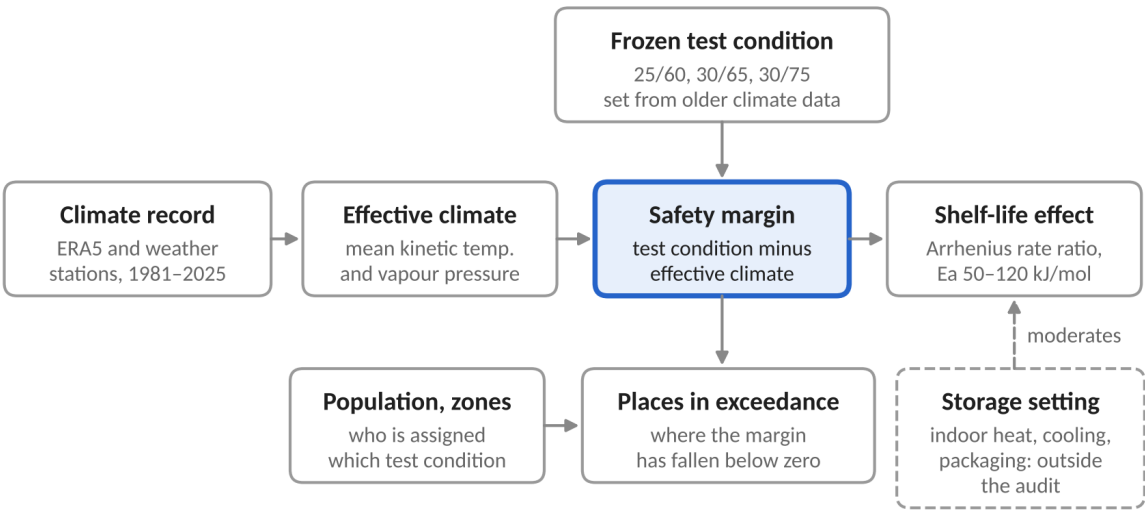


Figure 1. Conceptual model. A fixed test condition is compared each year with the effective climate; the difference is the safety margin. A negative margin shortens effective shelf life. How medicines are actually stored (indoors, cooled, packaged) moderates this link and is outside the audit.

3.1 Equations

Mean kinetic temperature over n equally weighted temperatures T_i (kelvin), with activation energy ΔH (default 83.144 kJ/mol) and gas constant R :

$$T_{MKT} = \frac{\Delta H/R}{-\ln\left(\frac{1}{n} \sum_{i=1}^n e^{-\Delta H/(RT_i)}\right)}$$

Temperature and humidity margins, where the test vapour pressure is the test relative humidity times the saturation vapour pressure e_s at the test temperature:

$$M_T = T_{test} - T_{MKT} \qquad M_P = RH_{test} e_s(T_{test}) - \bar{P}$$

Effective shelf life when the climate's MKT exceeds the test temperature (zero-order kinetics, activation energy E_a):

$$L_{eff} = L_{label} \exp\left[\frac{E_a}{R}\left(\frac{1}{T_{MKT}} - \frac{1}{T_{test}}\right)\right]$$

Computed reference values: the test vapour pressures are 19.0 hPa (25 °C / 60%), 27.6 hPa (30 °C / 65%), 29.7 hPa (30 °C / 70%) and 31.8 hPa (30 °C / 75%). At 30 °C, each extra degree of MKT speeds a typical reaction by 11.5% (6.7–16.9% for 50–120 kJ/mol), so a 24-month label behaves like 21.5 months at +1 °C and 19.3 months at +2 °C.

3.2 Hypotheses (fixed before the data run)

	Prediction	Rejected if
H1 Margin erosion	Effective temperature is higher in 2006–2025 than in 1981–2000	The 95% bootstrap interval of the difference includes zero
H2 Hidden exceedance	Some places already average above the test temperature assigned to their country	No place exceeds its condition in 2006–2025
H3 Humidity	Humidity margins erode more slowly than temperature margins	Humidity exceedance is more common than temperature exceedance
Positive control	Tokyo stays inside Japan's 25 °C condition, as found by Miura (2024)	Tokyo exceeds it — which would mean the method, not the climate, is wrong

3.3 What is new

- **Repeated measurement:** the condition is re-tested for every year from 1981 to 2025, not once.
- **Below country level:** places are compared with their country's condition, exposing hotter regions inside temperate-zone countries.
- **Two margins:** temperature and humidity are tracked separately, so the condition the climate breaks first is visible.
- **Built-in checks:** a positive control, an hourly-data bias check and a reanalysis-versus-station comparison.

4. Implementation Plan

The study is a secondary analysis of public climate data; it involves no human participants. Section 4.1–4.4 describe the method; Section 4.5 reports what the first run produced; Section 4.7 lists the steps that complete the full audit.

4.1 Data

- **ERA5 reanalysis** (Hersbach et al., 2020) for 10 cities, retrieved through the Open-Meteo historical API with the *era5* model: daily maximum and minimum 2 m temperature and daily mean dew point, 1981–2025.
- **NOAA GHCN-Daily station records** (Menne et al., 2012) for 8 places inside temperate-zone (25 °C) countries; years with fewer than 330 valid days were excluded.

- **NOAA Integrated Surface Database hourly records** (2024) for the method-bias check.
- **Test conditions** from WHO (2018): 25 °C for ICH-region countries (United States, Japan, Spain, Greece); 30 °C for Zone IV countries, with 65%, 70% and 75% RH variants for humidity.

4.2 Procedure

- Reconstruct a 24-hour temperature cycle from each day's maximum and minimum (cosine with peak at 15:00).
- Compute MKT per year with $\Delta H = 83.144$ kJ/mol and a 19 °C floor for heated indoor storage ("Grimm replica"); repeat without the floor and at 50 and 120 kJ/mol.
- Compute mean vapour pressure from daily mean dew point (Buck formula).
- Average each place over 1981–2000 and 2006–2025; estimate the difference with a 5,000-draw bootstrap over years; estimate the trend with the Theil–Sen slope.
- Count years above the test temperature in each period; convert exceedances to effective shelf life.

4.3 Decision rules

The hypotheses and rejection criteria in Section 3.2 were written before the data were downloaded. All results below are reported whether or not they support the hypotheses.

4.4 Validation

Method bias. Reconstructing MKT from daily maximum and minimum was compared with MKT from true hourly station data (Miami and Phoenix, 2024). The reconstruction reads about **0.3 °C high**, so all levels and exceedances below are reported after subtracting 0.3 °C. Period differences (H1) are unaffected, because the bias cancels in a difference.

Reanalysis versus station. Where both records cover the same years, ERA5 minus station MKT was: Athens +0.22 °C (21 yr), Delhi -0.67 °C (1 yr), Tokyo -0.46 °C (9 yr). Reanalysis tends to miss urban heat, so ERA5-based exceedances are, if anything, conservative. This validation set is small because NOAA holds few complete recent years for Indian and Japanese stations.

4.5 Execution status and first results

The first run was completed on 8 October 2026 for 18 places. Table 1 lists every result; Figures 2 and 3 show them.

Table 1. Bias-corrected mean kinetic temperature (°C) per period, change with 95% bootstrap interval, margin = test temperature minus 2006–2025 MKT (negative = climate above the test condition), and number of years above the test temperature. * = weather-station record.

Place	Source	Test	MKT 1981–2000	MKT 2006–2025	Change [95% CI]	Margin now	Years above (then → now)
Phoenix*	Station	25 °C	28.51	29.42	+0.91 [0.61, 1.23]	-4.42	20 → 20 of 20
Las Vegas*	Station	25 °C	26.27	27.49	+1.22 [0.88, 1.56]	-2.49	20 → 20 of 20
Dubai	ERA5	30 °C	30.48	31.81	+1.33 [1.02, 1.62]	-1.81	14 → 20 of 20
Tucson*	Station	25 °C	26.08	26.80	+0.72 [0.39, 1.07]	-1.80	19 → 20 of 20
Miami*	Station	25 °C	25.90	26.47	+0.57 [0.35, 0.81]	-1.47	19 → 20 of 20
Tampa*	Station	25 °C	24.96	25.67	+0.71 [0.45, 0.97]	-0.67	9 → 18 of 20
Houston*	Station	25 °C	24.68	25.41	+0.74 [0.44, 1.04]	-0.41	5 → 16 of 20
New Orleans*	Station	25 °C	24.04	24.92	+0.87 [0.58, 1.17]	+0.08	0 → 9 of 20
Seville*	Station	25 °C	23.60	24.35	+0.75 [0.44, 1.04]	+0.65	0 → 2 of 20
Nagpur	ERA5	30 °C	28.87	29.21	+0.35 [0.07, 0.62]	+0.79	0 → 1 of 20
Yangon	ERA5	30 °C	28.55	29.02	+0.47 [0.25, 0.67]	+0.98	0 → 0 of 20
Chennai	ERA5	30 °C	28.36	28.79	+0.43 [0.29, 0.57]	+1.21	0 → 0 of 20

Place	Source	Test	MKT 1981–2000	MKT 2006–2025	Change [95% CI]	Margin now	Years above (then → now)
Delhi	ERA5	30 °C	28.10	28.44	+0.35 [0.01, 0.68]	+1.56	0 → 0 of 20
Athens	ERA5	25 °C	22.72	23.34	+0.62 [0.40, 0.85]	+1.66	0 → 0 of 20
Colombo	ERA5	30 °C	27.34	27.94	+0.60 [0.36, 0.84]	+2.06	0 → 0 of 20
Mumbai	ERA5	30 °C	26.78	27.47	+0.69 [0.54, 0.83]	+2.53	0 → 0 of 20
Dhaka	ERA5	30 °C	26.73	27.45	+0.71 [0.53, 0.90]	+2.55	0 → 0 of 20
Tokyo	ERA5	25 °C	21.41	22.09	+0.68 [0.37, 1.02]	+2.91	0 → 0 of 20

H1 — margin erosion: supported in all 18 places. Effective temperature rose by 0.35 to 1.33 °C; every 95% interval excludes zero. Trends range from 0.10 to 0.47 °C per decade.

H2 — hidden exceedance: supported. 7 places average above their test temperature in 2006–2025: Phoenix, Las Vegas, Dubai, Tucson, Miami, Tampa and Houston. The clearest cases are on the US Gulf coast, inside an ICH region tested at 25 °C. Houston moved from 24.68 to 25.41 °C (5 → 16 of 20 years above 25 °C) and Tampa from 24.96 to 25.67 °C (9 → 18 years); in both, the climate crossed the line between the two periods. New Orleans averages 24.92 °C, on the line, and exceeded it in 9 of the last 20 years after none before. The desert cities of the US Southwest (Phoenix, Las Vegas, Tucson) were already above 25 °C in 1981–2000 and have moved further away. In Zone IV, Dubai averages 31.81 °C against 30 °C. Nagpur and Seville remain below on average but have begun to cross in individual years.

H3 — humidity erodes more slowly: rejected for humid coasts. Against 30 °C / 75% RH every Zone IV city keeps a humidity margin. Against 30 °C / 65% RH, Chennai (29.1 hPa), Colombo (29.9 hPa), Yangon (28.8 hPa) already exceed the chamber's 27.6 hPa; Colombo also exceeds the 30 °C / 70% chamber (29.7 hPa). Where a country tests at 30 °C / 65%, humidity — not heat — is the condition the climate breaks first. This echoes the reason WHO created Zone IVb; the contribution here is measuring it with current data.

Positive control: passed. Tokyo averages 22.09 °C against Japan's 25 °C condition, consistent with Miura (2024).

Shelf-life translation (default activation energy; range for 50–120 kJ/mol): a 24-month label behaves like 22.9 months in Houston (22.4–23.3); 22.3 months in Tampa (21.5–22.9); 20.4 months in Miami (18.9–21.7); 19.7 months in Dubai (18.1–21.3); 19.6 months in Tucson (18.0–21.3); 18.2 months in Las Vegas (16.1–20.3); 14.7 months in Phoenix (11.8–17.9). These figures describe the outdoor climate, not measured product exposure.

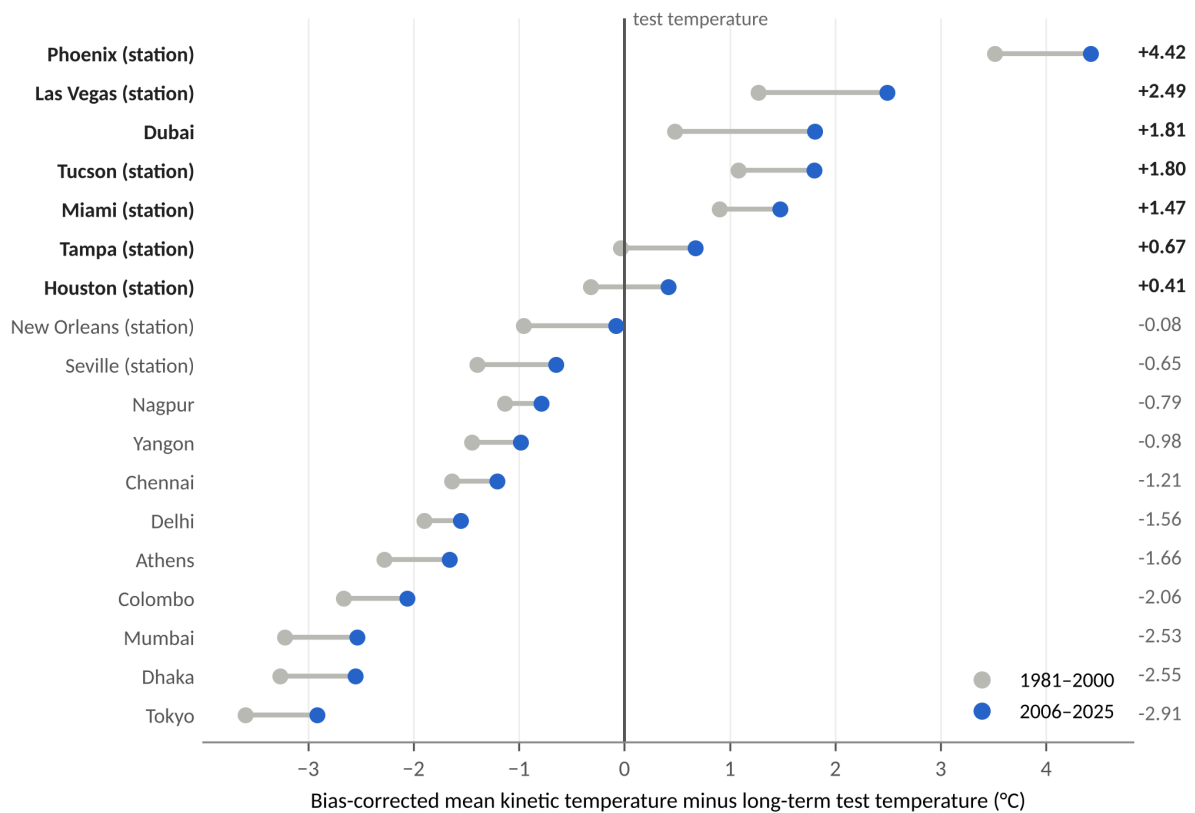


Figure 2. Every place moved toward or past its test temperature. Grey: 1981-2000; blue: 2006-2025; value at right: 2006-2025 margin (positive = above the test condition). Bias-corrected; station records marked.

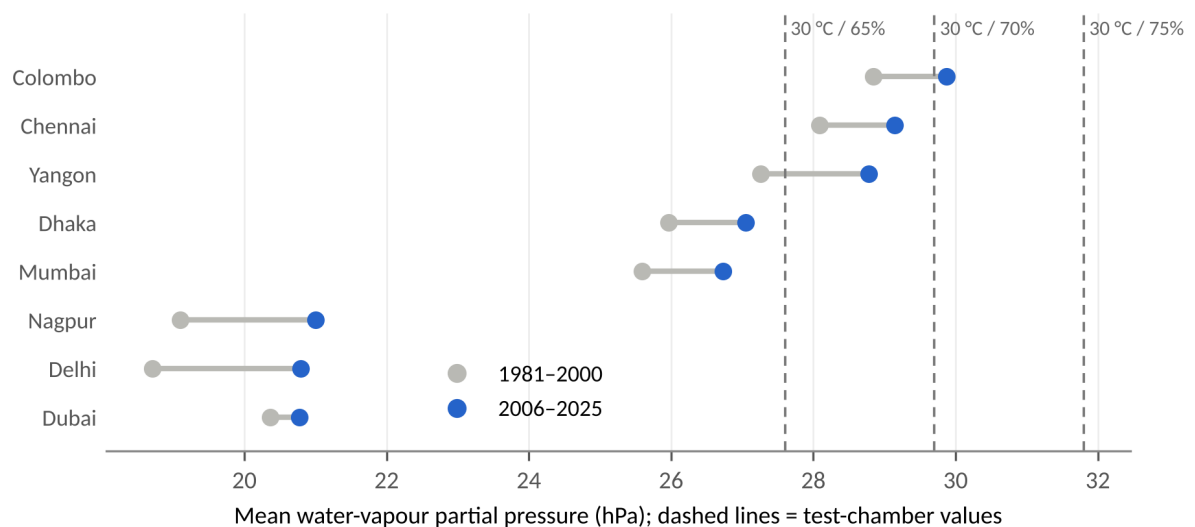


Figure 3. Water-vapour pressure in Zone IV cities against the three 30 °C test chambers. Coastal cities sit above the 65% chamber; none yet reaches the 75% chamber.

4.6 Limitations

- **Outdoor climate is not product temperature.** Many buildings in the United States and the Gulf are air-conditioned, and Grimm also used warehouse measurements. The results show that the test condition no longer bounds the outdoor climate it was meant to cover; they do not show that any product was exposed or failed.
- **Sample.** 18 places are not a population-weighted global sample. The ten ERA5 cities were chosen to span Zone II and Zone IV climates; the eight US and Spanish stations were deliberately chosen among the warmest large cities of 25 °C countries, to test H2. They show that exceedance exists, not how common it is.

- **Kinetics.** Arrhenius and zero-order assumptions do not fit every degradation pathway; results are reported across 50–120 kJ/mol.
- **Reconstruction.** Daily max/min reconstruction over-reads MKT by about 0.3 °C (corrected); ERA5 under-reads urban heat.
- **Zone assignments.** National conditions vary and change (India reportedly uses 30 °C / 70%); each country's current rule should be confirmed before policy use.

4.7 Next steps to complete the audit

- Run the gridded pipeline (Section 6) on ERA5 for South and Southeast Asia, weighted by Gridded Population of the World, to count people living where the margin is below zero.
- Extend validation with national meteorological station data (e.g. India Meteorological Department).
- Add projections (CMIP6) to estimate when each region's margin reaches zero.
- Pair with product-level evidence: temperature loggers in pharmacies and stability data for humidity-sensitive products.

5. References

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6. Prototype, Diagram & Visualization

Open audit tool. The analysis is a short, open Python pipeline (*stability_drift_pipeline.py*) with a self-test that reproduces the worked example used in ICH training material: temperatures of 20 °C and 40 °C give an MKT of 34.4 °C; the pipeline returns 34.36 °C. It downloads ERA5 data, computes MKT and vapour pressure per grid cell and year, and writes the margins against every test condition, ready for population weighting. The city-level results in this paper are released as *expired_by_climate_results.csv* (one row per place and year). Repository: [add public link].

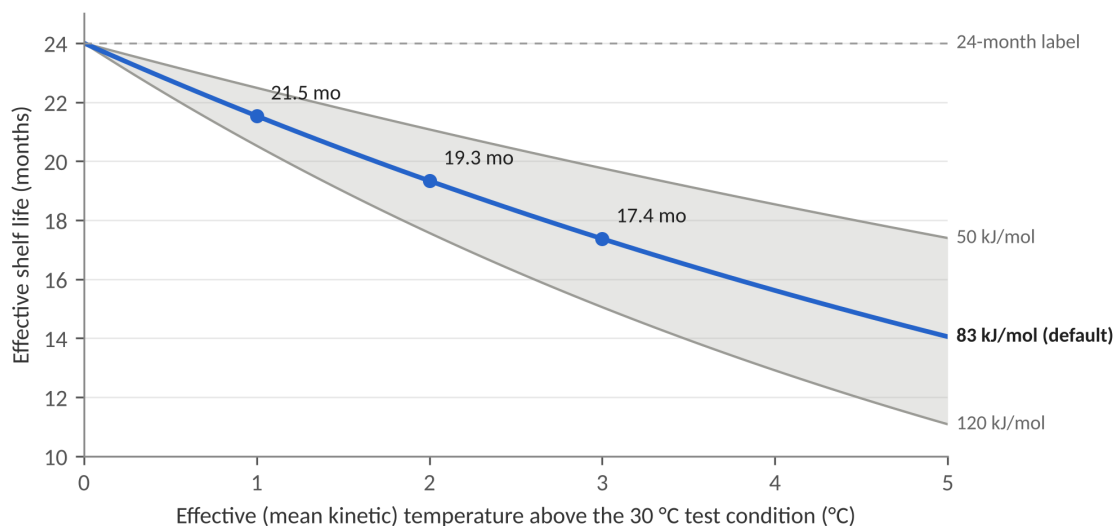


Figure 4. How much shelf life a lost degree costs. Effective shelf life of a product labelled 24 months at 30 °C when the climate's MKT is higher, for activation energies of 50–120 kJ/mol (band) and the default 83 kJ/mol (blue). Computed from the Arrhenius equation.

Reproducibility checklist. Data sources and access dates (Section 5); fixed periods, activation energies and decision rules (Sections 3.2, 4.3); bias correction value and how it was measured (Section 4.4); code with self-test; per-year output file. Anyone can re-run the audit for any city in minutes.